Harsh Kumar Jha

+91-9315956137 | contactharshjh@gmail.com | LinkedIn | GitHub | Portfolio

Skills

Languages	Python, Java, SQL
AI/ML	LLMs, Generative AI, Agentic AI, RAG, LLM Evaluation, Hugging Face Transformers, Sentence-Transformers, Scikit-learn, XGBoost, Red Teaming, Explainable AI (XAI)
LLM/AI	LLM Integration, AI Pipelines, AI Workflow Orchestration, Prompt Engineering, Model Evaluation, Tool Calling
Backend	FastAPI, REST APIs, JWT Authentication, Pydantic, SQLAlchemy
Databases	PostgreSQL, Redis, Supabase
DevOps	Docker, CI/CD, Railway, Vercel, Model Deployment, Monitoring
Developer Tools	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman

Experience

DomainLabs **Remote**
AI/ML Intern *June 2026 – Aug 2026*

- Conducted research in **AI safety**, adversarial testing, and **LLM evaluation** to assess model behavior, robustness, and reliability across diverse AI workflows.
- Designed and implemented **machine learning experiments, evaluation pipelines, and benchmarks** to measure LLM performance, robustness, and safety.
- Developed **adversarial threat taxonomies and red teaming workflows** for large language models, covering prompt-based attacks, failure modes, and model vulnerabilities.
- Contributed to **Bayora**, DomAIyn Labs' automated adversarial red teaming platform, through **LLM integration, model testing, and AI pipeline development**.
- Explored **model alignment, robustness, interpretability, and agentic AI workflows**, translating research findings into technical documentation, experiments, and actionable improvements.

MathonGo **Remote**
Research & Development Intern *Jun 2025 – Sep 2025*

- Validated **2,000+** Physics questions and structured/tagged educational datasets for JEE/NEET question banks, applying AI-assisted quality assurance to improve data quality and retrieval efficiency.

Projects

Hallucination Hunter | *Python, FastAPI, DeBERTa-v3, Sentence-Transformers, spaCy, Docker* GitHub

- Engineered a claim-level **LLM evaluation and hallucination-detection pipeline** using DeBERTa-v3 NLI and Sentence-Transformers to verify generated outputs against source context.
- Built **production-ready REST APIs** with explainable confidence scoring, alignment heatmaps, health checks, and Dockerized deployment for direct integration with **RAG pipelines**.

AppForge AI | *Python, FastAPI, Next.js, Pydantic, SQLAlchemy, Cerebras, Groq, OpenAI* GitHub

- Built an **LLM-powered multi-stage agent pipeline** that converts natural-language prompts into applications through intent extraction, system design, schema generation, validation/repair, and execution checks.
- Integrated **OpenAI, Groq, and Cerebras** through a primary-fallback routing layer with automated repair loops, token-budget controls, and end-to-end **AI workflow orchestration** deployed on Railway.

FrameOS | *Next.js, Node.js, PostgreSQL, Supabase, Redis, Docker* GitHub

- Engineered an **AI-native workflow orchestration platform** automating research, script generation, rendering, publishing, and analytics through distributed, event-driven pipelines.
- Implemented **Redis-based job queueing, JWT authentication, and Dockerized deployment** to support reliable multi-step execution and scalable autonomous content workflows.

Education

Maharaja Surajmal Institute of Technology, New Delhi 2025 – 2029
B.Tech, Electronics and Communication Engineering **CGPA: 8.36/ 10**